

Apex Essentials

The server-side language of the Salesforce Platform – syntax, DML, triggers, and governor limits.

• Primitive Types

- Integer, Long, Double, Decimal
- String, Boolean, Id
- Date, Datetime, Time
- Blob, Object

```
Decimal price = 19.99;
Date today = Date.today();
Id acctId = '001...';
```

• Collections

```
List<String> names = new List<String>
{'a','b'};
Set<Id> ids = new Set<Id>();
Map<Id, Account> byId =
  new Map<Id, Account>([SELECT Id FROM
Account]);
```

- List – ordered, allows duplicates.
- Set – unique, unordered.
- Map – key/value; great for lookups by Id.

• DML Statements

```
insert acct;
update contactList;
upsert leadList External_Id__c;
delete oldRecords;
```

- Operate on **lists**, not single rows, to bulkify.
- Database.insert(list, false) → partial success + SaveResult[].

• SOQL in Apex

```
List<Account> accts =
[SELECT Id, Name FROM Account
WHERE Industry = :ind LIMIT 50];

for (Account a : [SELECT Id FROM Account]) {
  // SOQL for-loop chunks 200 rows – heap-
safe
}
```

• Trigger Skeleton

```
trigger AccountTrigger on Account
(before insert, after update) {
  if (Trigger.isBefore && Trigger.isInsert)
  {

AccountHandler.onBeforeInsert(Trigger.new);
  }
}
```

- One trigger per object; delegate to a **handler class**.
- Context: Trigger.new, .old, .newMap, .oldMap.

• Trigger Context Vars

VAR	AVAILABLE
isBefore/isAfter	all events
isInsert/Update/Delete/Undelete	matching event
new / newMap	insert, update, undelete
old / oldMap	update, delete
size	all events

• SOSL & Exceptions

```
try {
  insert acct;
} catch (DmlException e) {
  System.debug(e.getMessage());
} finally {
  // cleanup
}
```

- Custom: class MyException extends Exception {}
- throw new MyException('bad');

• Async Apex

- @future(callout=true) – fire-and-forget method.
- Queueable – chainable, supports state.
- Batchable – process millions of rows in chunks.
- Schedulable – run on a cron schedule.

```
System.enqueueJob(new MyQueueable());
```

• Governor Limits

Per synchronous transaction:

RESOURCE	SYNC	ASYNC
SOQL queries	100	200
DML statements	150	150
Rows via DML	10,000	10,000
CPU time	10,000 ms	60,000 ms
Heap size	6 MB	12 MB
Callouts	100	100

Bulkify: no SOQL/DML inside loops. Query once, work with collections.

• Testing

```
@isTest
private class AccountHandlerTest {
  @isTest static void insertsOk() {
    Test.startTest();
    insert new Account(Name='T');
    Test.stopTest();
    System.assertEquals(1,
      [SELECT COUNT() FROM Account]);
  }
}
```

- 75% org coverage required to deploy.
- Use @testSetup; never rely on org data (SeeAllData=false).

• Security

```
List<Account> a = [SELECT Name FROM Account
WITH USER_MODE];
Account clean = (Account) Security
.stripInaccessible(AccessType.READABLE,
recs)
.getRecords()[0];
```

- WITH USER_MODE enforces CRUD + FLS + sharing.
- API v67+: DB ops run in user mode by default and classes default to with sharing; WITH SECURITY_ENFORCED is removed – use WITH USER_MODE.
- with sharing / without sharing / inherited sharing on classes.